

CHAPTER 6 - LIFE PROCESSES IN PLANTS

6.1 Nutrition in Plants

All living organisms need food to survive. Food gives organisms the energy they need to grow, repair damaged parts and carry out life processes. Plants are different from animals in one important way: green plants make their own food. Organisms that make their own food are called autotrophs.

6.2 What Is Photosynthesis?

Photosynthesis is the process by which green plants and some other organisms use sunlight to synthesise food from carbon dioxide and water. The word itself explains the process: "photo" means light, and "synthesis" means putting together. So photosynthesis literally means putting together with light.

In green plants, photosynthesis takes place in tiny structures inside the leaf cells called chloroplasts. Chloroplasts contain a green pigment called chlorophyll. Chlorophyll absorbs light energy from the sun and converts it into chemical energy. This is why most leaves appear green.

6.3 Raw Materials

Three things are required for photosynthesis to happen:

1. Sunlight - the source of energy. Without light, the process stops.
2. Carbon dioxide - taken in from the air through tiny pores called stomata.
3. Water - absorbed from the soil by the roots and carried up to the leaves.

It is important to understand that the plant does not eat the soil. The soil supplies water and dissolved minerals only. The food itself is manufactured in the leaves. Minerals such as nitrogen and magnesium help the plant build chlorophyll and proteins, but they are not food.

6.4 Products of Photosynthesis

During photosynthesis, the plant produces glucose, which is a simple sugar. The glucose is the food. Some of it is used immediately for energy, and the rest is converted into starch and stored in the leaves, stems, roots, fruits and seeds. This is why a potato tastes starchy and why seeds are rich in stored food.

Oxygen is released as a by-product of photosynthesis. This oxygen passes out of the leaf through the stomata into the air. Almost all the oxygen in the atmosphere that animals and humans breathe was produced by photosynthesis.

6.5 The Word Equation

Carbon dioxide + Water $\xrightarrow{\text{(sunlight, chlorophyll)}}$ Glucose + Oxygen

6.6 Why Photosynthesis Matters

Photosynthesis is the foundation of nearly every food chain on Earth. Plants are called producers because they produce food. Animals that eat plants are called consumers. When you eat a chapati, the energy in it originally came from sunlight, captured by a wheat plant through photosynthesis.

Photosynthesis also removes carbon dioxide from the atmosphere and releases oxygen, which helps keep the balance of gases in the air stable.

6.7 Points to Remember

- Photosynthesis occurs mainly in the leaves, in the chloroplasts.
- Chlorophyll traps light energy and converts it into chemical energy.
- Raw materials: carbon dioxide, water and sunlight.
- Products: glucose (food) and oxygen (by-product).
- Soil is not food for a plant; it supplies water and minerals.
- Plants are autotrophs and producers.